



VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute | Accredited by NAAC with 'A' Grade
Gokulam 3rd stage, Mysuru - 570002



INFOTHON



THEME: SOCIAL INNOVATION

TEAM NAME: DEV SQUAD

PROBLEM STATEMENT:



STUDENT INFORMATION AND PERFORMANCE TRACKING SYSTEM:

OBJECTIVE: TO DEVELOP A SECURE, USER-FRIENDLY STUDENT INFORMATION SYSTEM (SIS) THAT PROVIDES STUDENTS AND PARENTS WITH REAL-TIME ACCESS TO ATTENDANCE RECORDS, ACADEMIC PERFORMANCE (CIE AND SEE GRADES), ACHIEVEMENTS, AND COUNSELING REPORTS.

SUMMARY: EMPOWERED AIMS TO REVOLUTIONIZE THE WAY STUDENTS AND PARENTS ENGAGE WITH ACADEMIC INFORMATION BY CREATING AN INTEGRATED PLATFORM THAT ADDRESSES KEY AREAS OF STUDENT PERFORMANCE AND WELL-BEING. CURRENT SYSTEMS OFTEN LACK THE SEAMLESS INTEGRATION AND ACCESSIBILITY NEEDED FOR EFFECTIVE COMMUNICATION AND TRACKING. THIS PROJECT SEEKS TO FILL THAT GAP BY PROVIDING A COMPREHENSIVE, REAL-TIME SIS THAT ENSURES TRANSPARENCY, ENCOURAGES STUDENT ENGAGEMENT.

PROPOSED SOLUTION:

(WEB APPLICATION)



THE PROPOSED SOLUTION AIMS TO CREATE A COMPREHENSIVE AND USER-FRIENDLY STUDENT INFORMATION AND PERFORMANCE TRACKING SYSTEM BY IMPLEMENTING KEY FEATURES TAILORED TO EACH ROLE. STUDENTS CAN ACCESS THEIR OWN RECORDS, TEACHERS CAN MANAGE RECORDS AND UPLOAD MATERIALS, PARENTS CAN MONITOR THEIR CHILD'S PROGRESS, AND ADMINS OVERSEE THE ENTIRE SYSTEM, INCLUDING COURSE MANAGEMENT. THE SYSTEM ENSURES SECURE LOGIN WITH ROLE-BASED ACCESS AND PERSONALIZED DASHBOARDS FOR EACH USER. FEATURES LIKE REAL-TIME ATTENDANCE TRACKING, ACADEMIC PERFORMANCE MONITORING, AND ONE-TO-ONE VIDEO CALLING FOR PARENT-TEACHER MEETINGS ENHANCE COMMUNICATION AND TRANSPARENCY. ADDITIONALLY, FEEDBACK TOOLS, DATA ANALYTICS, MOBILE ACCESSIBILITY, AND ROBUST SECURITY MEASURES MAKE EMPOWERED AN EFFECTIVE AND RELIABLE PLATFORM FOR ALL USERS.

FEATURES AND IMPACT:



- **ROLES AND RESPONSIBILITIES ARE DIVIDED AS FOLLOWS:** STUDENTS CAN ACCESS THEIR OWN RECORDS, TEACHERS CAN MANAGE STUDENT RECORDS AND UPLOAD MATERIALS, PARENTS CAN MONITOR THEIR CHILD'S PROGRESS, AND ADMINS OVERSEE THE ENTIRE SYSTEM LIKE ADDING AND DELETING THE COURSE ETC.
- **USER AUTHENTICATION AND ROLE-BASED ACCESS:** SECURE LOGIN FOR STUDENTS, PARENTS, TEACHERS, AND ADMINS WITH DIFFERENT LEVELS OF ACCESS PERMISSIONS.
- **INTEGRATION CAPABILITIES:** ABILITY TO INTEGRATE WITH OTHER EDUCATIONAL PLATFORMS, DATABASES, OR SYSTEMS USED BY THE INSTITUTION.



- **DASHBOARD:** PERSONALIZED DASHBOARD FOR EACH USER ROLE DISPLAYING RELEVANT INFORMATION LIKE DETAILS ETC.
- **ATTENDANCE SYSTEM:** REAL-TIME ATTENDANCE RECORDS FOR STUDENTS, AND FOR PARENTS ON ABSENTEEISM.
- **ACADEMIC PERFORMANCE TRACKING:** DISPLAYING CONTINUOUS INTERNAL EVALUATION (CIE) AND SCHOOL END EXAMINATION (SEE) GRADES.SEMESTER-WISE OR TERM-WISE PERFORMANCE SUMMARIES.
- **PARENT TEACHERS MEETING:** ONE TO ONE VIDEO CALLING FEATURE FOR PARENT AND FACULTY WHERE THEY CAN DISCUSS THE REPORT OF THE STUDENT.



- **FEEDBACK AND SURVEYS:** TOOLS FOR STUDENTS, PARENTS, AND TEACHERS TO PROVIDE FEEDBACK AND PARTICIPATE IN SURVEYS.
- **DATA ANALYTICS AND REPORTING:** GENERATE REPORTS ON ATTENDANCE, ACADEMIC PERFORMANCE TRENDS, AND OTHER METRICS FOR ADMINISTRATORS AND TEACHERS TO ANALYZE.
- **MOBILE ACCESSIBILITY:** RESPONSIVE DESIGN OR DEDICATED MOBILE APP FOR EASY ACCESS ON SMARTPHONES AND TABLETS.
- **SECURITY AND PRIVACY:** ENSURE DATA ENCRYPTION, SECURE AUTHENTICATION PROTOCOLS, AND COMPLIANCE WITH DATA PROTECTION REGULATIONS.

TECH STACK AND LIBRARIES:

INFOTHON



WEB RTC: FOR VIDEO CALLING



TAILWIND CSS